

Ashutosh Sharma

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Professional Summary

Detail-oriented Mechanical Engineer (EIT) with strong background in process engineering, CFD, thermal systems, and multidisciplinary project delivery. Experienced in design calculations, process modeling, technical documentation, and cross-functional collaboration. Skilled in applying simulation tools, optimization techniques, and data-driven problem-solving to improve system performance.

Work Experience

Project Intern – Flowthermolabs

Jul 2024 – Jul 2025 | Remote

- Assisted in the design and development of thermal management solutions for electronic devices, contributing to the successful completion of 3 key projects.
- Conducted detailed research and analysis on thermal dynamics, leading to a 15% improvement in the efficiency of cooling systems used in prototypes.
- Collaborated with a team of 5 engineers to troubleshoot and resolve technical issues, ensuring project milestones were met on time and within budget.
- Prepared and presented technical reports and presentations, enhancing project documentation and communication with stakeholders by 20%.

Assistant Research Fellow – Nepal Academy of Science & Technology

Nov 2021 – Nov 2022 | Lalitpur | Nepal

- Optimized process parameters for Titanium Dioxide Nanoparticle synthesis for Dye-Sensitized Solar Cells, achieving a 20% increase in energy conversion efficiency.
- Conducted XRD and FTIR Characterization techniques, improving the accuracy of nanoparticle characterization.
- Utilized SolidWorks to design and Ansys to perform detailed stress analysis on Francis Turbine components, identifying critical stress points and enhancing design precision by 30%.
- Applied DFMEA using Microsoft Project, systematically evaluating potential failure modes, leading to a 15% increase in turbine reliability.
- Conducted design verification by reviewing reports, calculations, and drawings, ensuring compliance with ASME and CSA N285.0 codes.

Assistant Mechanical Engineer – Sufal Pvt. Ltd.

Aug 2020 – Sept 2021 | Lalitpur | Nepal

- Managed project timelines and deliverables for 5+ engineering projects, ensuring 100% on-time completion and adherence to client specifications.
- Led a cross-functional team of 10 engineers and technicians, improving team efficiency by 20% through effective leadership and project coordination.
- Implemented cost-saving measures, reducing project costs by 15% without compromising on quality or performance.
- Provided engineering support in preparing technical deliverables, including specifications, test procedures, and design submissions equipment components, enhancing project compliance with regulatory standards.
- Utilized project management software to track project progress and resources, leading to a 10% improvement in project scheduling accuracy.

Project Intern – Non-Ferrous Materials Technology Development Centre (NFTDC)

Jun 2019 – Jul 2020 | Hyderabad | India

- Characterized materials using advanced techniques, including SEM and Energy Dispersive X-ray Spectroscopy (EDS), improving analysis accuracy by 25% and enhancing the understanding of material composition.
- Utilized Horiba Particle Size Analyzer to determine particle distribution, achieving a 95% precision rate in measurement, which facilitated more accurate and reliable research results.
- Prepared surfaces for Optical Microscopy, resulting in a 30% improvement in surface morphology analysis using Taylor Hobson Profilometer, leading to more detailed and accurate observations.
- Conducted Magnetron Sputtering (DC & RF) for thin film sensor development on turbine blades, increasing sensor efficiency by 15% and improving the overall performance of the turbine blades.
- Coordinated large-scale projects of high technical and commercial complexity, ensuring seamless interface between disciplines and adherence to project timelines.

Engineer Trainee – Sufal Pvt. Ltd.

Aug 2017 – Aug 2018 | Lalitpur | Nepal

- Interpreted engineering drawings and technical manuals accurately, ensuring 100% adherence to project specifications and design requirements.
- Executed conceptual and detailed design activities for mechanical systems, such as equipment sizing, component selection, and layout, resulting in optimized operations.

- Coordinated Bill of Materials (BOM) and inventory management with outside suppliers, improving spare parts availability by 20% and enhancing equipment reliability.
- Implemented a quality management system (ISO9001, ISO13485), ensuring compliance with industry standards and achieving a 95% reduction in defects.
- Designed and modified molds, generating G codes for CNC machining with high proficiency, resulting in a 30% improvement in production efficiency.

Education

Fanshawe College – Canada	Aug 2023
<i>Post Graduate Certificate (Applied Mechanical Design)</i>	
National Institute of Technology Warangal – India	Aug 2020
<i>MTech (Materials & System Engineering Design)</i>	
M.S. Ramaiah Institute of Technology – India	Aug 2017
<i>B.E. (Mechanical Engineering)</i>	

Certifications

Engineer-in-Training (EIT) – Professional Engineers Ontario (PEO)	
Registered Mechanical Class “A” Engineer – Nepal Engineering Council	
Material Behavior – Georgia Institute of Technology (Coursera)	Sept 2021 – Sept 2021
Python Data Structures – University of Michigan (Coursera)	Jul 2022 – Jul 2022
Machine Learning – Stanford University (Coursera)	Feb 2022 – Feb 2022
A Hands-on Introduction to Engineering Simulations – CornellX (Cornell University)	Aug 2021 – Aug 2021

Skills & Interests

Technical Knowledge: CAD (SolidWorks, Siemens NX, CATIA, Autodesk Inventor, PTC Creo), CAE Software (ANSYS), Programming Languages (MATLAB, Python, C), Simulation of Machining Processes using CAM Packages (ESPRIT), Microsoft Office, LaTeX, Mechanical Design and Engineering, Equipment Sizing and Selection	
Professional Skills: Project Management, Research and Development, Technical Documentation, Process Optimization, Quality Assurance, Compliance, Inventory Management, Budget Management	
Soft Skills: Problem-solving, Team Collaboration, Communication (verbal and written), Leadership, Adaptability	

Projects

Design Optimization of Electric Vertical Takeoff and Landing Drone (EVTOL)	May 2023 – Aug 2023
• Optimized the design of an EVTOL drone for intercity purposes, achieving a 15% reduction in weight and a 10% increase in flight efficiency.	
Optimization of Process Parameters for Titanium Dioxide Nanoparticles	Nov 2021 – Nov 2022
• Produced TiO2 nanoparticles for dye-sensitized solar cells using the Sol-Gel method, enhancing production efficiency by 20% through Design of Experiments.	
Design Improvisations for IT SOFC	Aug 2019 – Jul 2020
• Fabricated and optimized a Solid Oxide Fuel Cell stack on SS-316 using screen printing, plasma glazing, and ink printing, improving performance by 25%.	
Finite Element Analysis Using MATLAB	Dec 2018 – Dec 2018
• Conducted stress analysis on beam and truss structures using MATLAB, achieving a 15% improvement in accuracy for stiffness matrix, displacement vectors, and support reactions.	
Wear Behavior of Rice Husk Ceramic Composites	Apr 2018 – Apr 2018
• Studied the strength of rice husk composites, achieving 10% higher strength compared to individual components, contributing to sustainable green construction materials.	
Optimizing Ambulance Path for Blood Supply During Kerala Flood	Apr 2019 – Apr 2019
Design and Fabrication of Paddy Transplanter	Jan 2017 – Jun 2017